

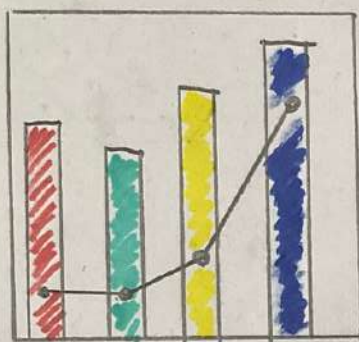
① IDEAS

- Bar chart (lined) comparing population growth to population density over time

Melbourne map



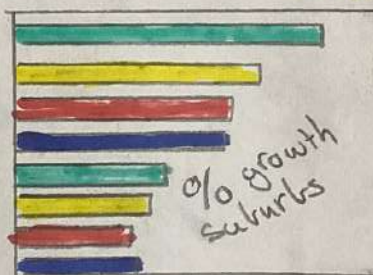
- Point map of built up area/year
- Heatmap of built up area/year
- Choropleth of population density per year



(No red green though obviously)

Good at corresponding population density to built up areas

Alternatively... Areachart?



② FILTER

Point map & Heatmap cannot go together - it's one or the other
Again, either area chart or line/bar chart.
Need zoom/pop years slider

1975 2030

③ CATEGORISE

1. Built up area map good for non interactive temporal

↓

2. Pop-density Choropleth

↓

3. Density type & comparison to built up

↓

4. Government initiatives and the temporal impact

④ COMBINE & REFINES

Horizontal bar chart & Choropleth map - two different, informative methods of displaying data.

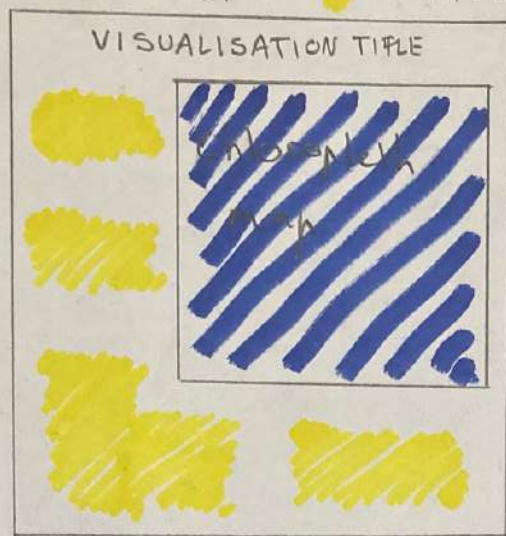
Want to place emphasis on the storytelling element of this vis - aka provide insight into the Urban rural divide. Compare different city urban areas population density e.g. Singapore, Melbourne etc.
This could replace stacked bar chart?

⑤ QUESTION

Does this satisfy what the client wants? Yes → it's a storytelling rather than simply a collection of informative tidbits,

① LAYOUT

Static Background image



Fit

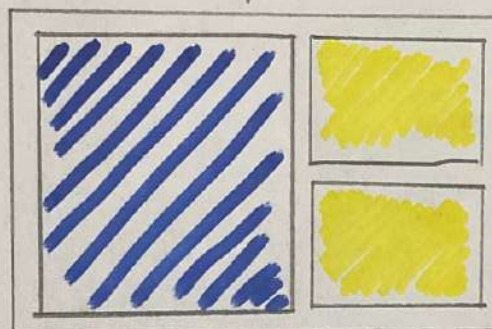


Map/Chart



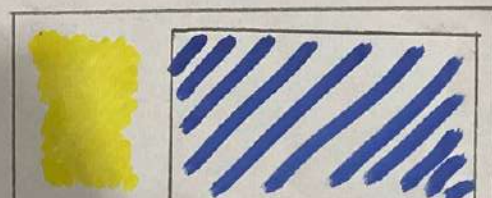
Text/Other

1080px



← Chloropleth about Melbourne Suburb Population Density over time

← Population Density Area chart



← TBD Probably More Density Stuff

⑤ INFORMATION

Sheet No. = 2

Author = Blake Travers

Date = 21-10-2025

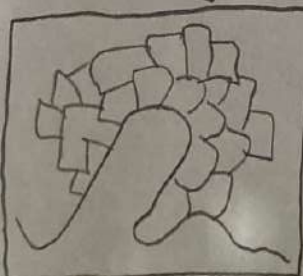
Title = HTML Layout

③ OPERATIONS

1. Scroll vertically between main boxes
 2. Tooltip for all data
 3. Chloropleth map filter by year every 5 years within data date range 1970-2030 (projected 2025 & 2050)
 4. Table data dropdown?
- Purpose is storytelling so maybe not

② FOCUS

Focus on WHITESPACE! With Background & Figure ground must use adequate whitespace effectively.



Interactive Chloropleth Map could be the largest idiom, Most important for domain?

Ensure all colour schemes are sequential for pop density diverging is not appropriate

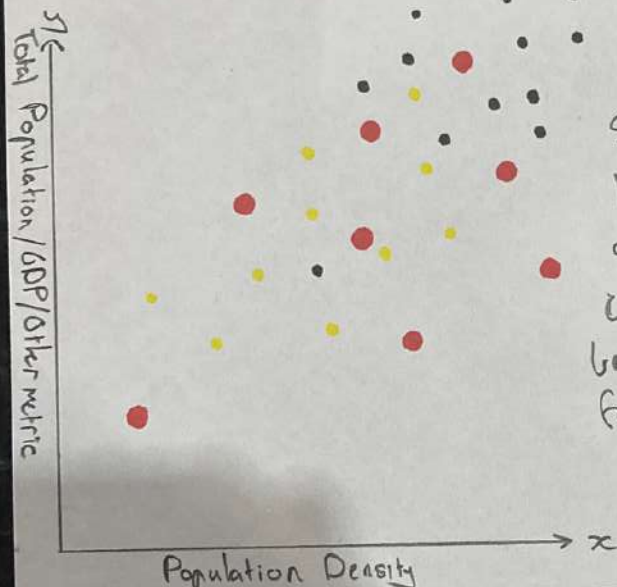
④ DISCUSSIONS

Need to reformat Raster data to fit suburbs. Use TOPOSSON polygons through Mapshaper, and find all points within

① LAYOUT



→ Colour by region
 → Focus on Melbourne
 → ID graph - there is not much going on here! For the selected audience we can afford to make a little more complex



⑤ INFORMATION

Sheet No. = 3

Authors = Blake Travers

Date = 21-10-2025

Title = HTML Layout + Visualisation idoms

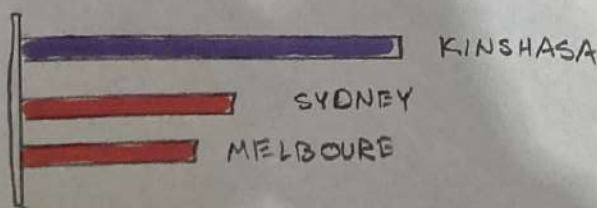
③ OPERATIONS

Pan ~~late~~ density / other metric in scatter plot to allow user to view at different granularity.

For Horizontal bar chart - can select each ^{city} ~~country~~ for more information, perhaps vertically scroll to see more cities, or select from a list of cities to be displayed.

② FOCUS

Focus needs to not be on the amount, or a number of data that can be displayed, but the quality of what the user is viewing. i.e. Less is better with at least regards to the scatter / horizontal plan.

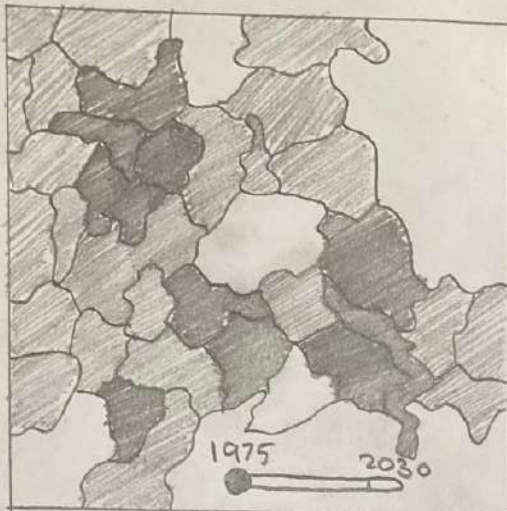


④ DISCUSSIONS

Pros of Bar chart
 → Simple, easy to understand
 → Low Data-Interaction
 Doesn't convey a lot of information while still wasting a lot of ink.

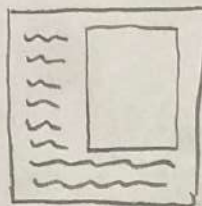
Pros of Scatter plot
 → High Data Ink ratio
 Clean, simple, and uses whitespace more effectively than Bar chart

① LAYOUT



Suburb-
Chloropleth
map with
tooltip:

Name =
Population =
P. Density =
Year =



↳ Embedded
like so
Sequential, single
hue colour
scheme!

↳ Same colour
Scheme as
area chart
for user
experience

Area chart explores the same
domain & attributes as chloropleth
map → but BINNED.



② FOCUS

Categories:

- 0-100 → Rural
- 101-500 → Periurban
- 501-2499 → Suburban
- 2500-10000 → Urban
- 10000+ → City Centre

Colour Schemes - THE SAME! Extremely important to
ensure audience knows that the message is the same

Focus on comparison
these idioms make. They
go hand-in-hand with
each other and should
be used to collectively
tell a story like what's
in the specifications.

⑤ INFORMATION

Sheet NO. = 4

Author = Blake Travers

Date = 22-10-2025

Title = HTML Layout &
containers

③ OPERATIONS

Yearly slider

Change YEAR to determine
what year of data is viewed.
→ Good for location comparison

Hover over suburbs

Display specific Year/bin
population / Suburb location
→ Display suburb intrinsic?

Hover over area chart

Highlight specific Year/bin
combo
→ Great for time comparison
→ Fills in the holes left by
chloropleth map filter

④ DISCUSSIONS

Need to decide how
each suburb will be
colour coded.

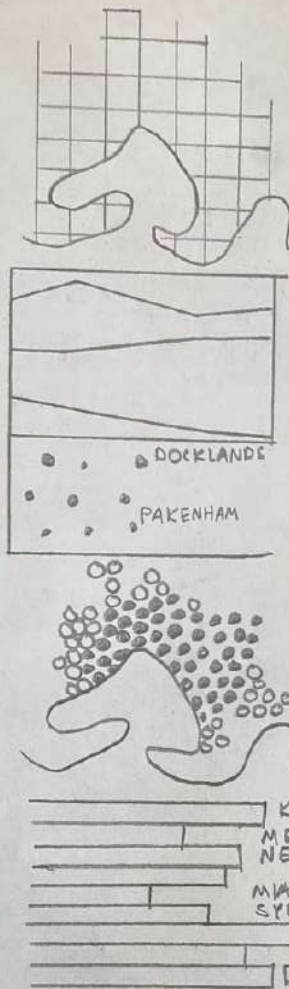
Ensure suburb tooltip does
not have too much

Add complexity to area charts
very simple at the moment,
can spice it up to match
its partner chloropleth.

LAYOUT

TITLE
CHLORO PLETH MAP + Population Density Temporal
AREA MAP → See sheet 4 for differences in purpose & this and chloropleth
SCATTER PLOT Population growth per suburb

POINT MAP Built up area by location → shows difference between industrial & residential
HORIZONTAL BAR Comparison between cities
Conclusion & Meta



Talk about
Urban sprawl

INFORMATION

Sheet No. 5

Author = Blake Travers

Date = 26/10/2025

Title = Final Design Sheet

OPERATION

Vertical scroll to
pan between idiom

→ Keep in mind max
1080px width!

Click on points to
highlight similar for
Horizontal Bar & Scatter
Plot

→ No need to do it for
area map

→ Won't be able to do it
for scatter plot due to
how it works & data size

FOCUS

Story following the following steps

1. What is urban sprawl?
2. How does urban sprawl apply to Melbourne
3. How has Melbourne counteracted urban sprawl
4. How does Melbourne's urban sprawl compare to that of other cities?

Focus on STORYTELLING PROVIDING A
Focus on CLEAR FLOW

DETAIL

Next steps

1. Setup Svealite
Json files → one for
each idiom

2. Source data

→ Population & Built-up

→ Suburb boundaries
(most important) TOPOJSON

→ City densities